time resolutions, namely $time\_ad1$, $time\_ad2$, and $time\_ad3$. Subsequently, a convolution with a sufficiently long window is applied to the shortest temporal feature sequence $time\_ad3$, using zero padding at both ends as anchors to encode the relative temporal position of each token in the sequence. Then, we progressively upsample the temporal feature sequences from short to long, using residual connections to preserve temporal features at different scales. Finally, the temporal feature sequence $time\_ad\_out$ is restored to the same length as the video features after token shuffling and aligned in the channel dimension through a linear layer.

B INSTRUCTION-TUNING DATA

We fine-tuned VideoChat-T using 432K data, which includes 349K instances from TimePro and 82K instances from normal data. All videos were sampled from existing open-source video datasets, with specific information about the relevant data provided in Table 6.

Set	Task	Sourse	Instance Num
TimePro	Temporal Video Grounding	DideMo	32,944
		QueryD	14,602
		HiREST-grounding	459
	Dense Video Captioning	ActivityNet-Captions	10,009
		ViTT	5,086
		Youcook2	8,700
	Video Summarization	TVSum	50
		SumMe	25
	Step Localization and Captioning	COIN	9,026
		HiREST-step	459
	Transcribed Speech Generation	YT-Temporal	31,190
	Reasoning Temporal Localization	ActivityNet-RTL	33,557
	Multi-format Temporal Grounding	InternVid-VTime	100,000
	Highlight Detection	ActivityNet-HL	10,340
	Temporal Grounded Caption	CosMo-TGC	93,118
Normal	Conversation	VideoChatGPT	13,303
		VideoChat	13,884
	Video QA	EgoQA	7,813
		MovieChat-QA	808
	Reasoning	STAR	45,731
	Caption	MovieChat-Caption	808

Table 6: The complete instruction fine-tuning data used for training. We utilized a total of approximately 432K data points, which can be divided into 349K instances of TimePro and 82K instances of regular video data, covering 13 tasks across 21 datasets.

We evaluate the quality of the data from three perspectives: diversity, length, and difficulty. We strive to include different datasets for various tasks, and the distribution of videos in the datasets is as broad as possible. The length of the videos should be controlled within an appropriate range, as excessively long or short videos may pose challenges for training. Each query should clearly describe the video content of the target time segment and avoid corresponding to multiple time segments in the video. Based on these principles, we have screened and integrated existing high-quality datasets, which significantly contribute to enhancing the model's temporal awareness capabilities.

TimePro encompasses a series of open-source temporal grounding datasets that we have integrated, cleaned, and refined, such as TimeIT (Ren et al., 2024), ANet-RTL (Huang et al., 2024b), and InternVid-VTime (Huang et al., 2024a). These high-quality open-source datasets have been experimentally validated by us. We also added two new self-made datasets, ANet-HL and CosMo-TGC.

Temporal Video Grounding. This task involves providing a natural language query and requires outputting the corresponding video's start and end times. The datasets include DiDeMo (Anne Hendricks et al., 2017), QuerYD (Oncescu et al., 2021), and HiREST-grounding (Zala et al., 2023), aiming to achieve precise temporal localization during user interaction with natural language.

Dense Video Captioning. This task requires the model to detect a series of events occurring in a given video and output the corresponding timestamps and coarse-grained descriptions. The datasets for this part include ActivityNet-Caption (Krishna et al., 2017), ViTT Huang et al. (2020), and YouCook2 (Zhou et al., 2018), which help the model learn the temporal relationships between different events within the video.

Video Summarization. The goal of this task is not to summarize at the semantic level of natural language, but to determine a set of compressed frames or clips in the form of timestamps, representing the most informative content in a given video. Our datasets include TVSum (Song et al., 2015) and SumMe (Gygli et al., 2014), which effectively combine the model's temporal perception capabilities with its semantic content inference abilities.

Step Localization and Captioning. This task differs from dense video captioning as it is designed to segment and describe the important steps within a long video. We have integrated two datasets, COIN (Tang et al., 2019) and HiREST-step (Zala et al., 2023), which can help the model learn the procedural temporal logic relationships of different steps within a single event.

Transcribed Speech Generation. The purpose of this task is to predict speech content and its corresponding start and end timestamps based on visual signals in the video. Including the YT-Temporal (Zellers et al., 2022) dataset, this task can be viewed as a weakly supervised event localization and description task.

Reasoning Temporal Localization. The answers to the questions in this task include both timestamps and explanations. We used the ANet-RTL (Huang et al., 2024b) dataset as training data for this task. By combining temporal localization and reasoning, we can more specifically enhance the model's temporal perception capabilities.

Multi-format Temporal Grounding. This task includes both single-turn and multi-turn dialogues, with a variety of question types. We use the InternVid-VTime (Huang et al., 2024a) dataset for training this task. The broader range of task types and more diverse output formats can effectively enhance the model's temporal generalization capabilities.

Highlight Detection. Unlike video summarization, this task identifies only the most salient moments of a video in response to a natural language query, without covering the entire scope of the original video (Lei et al., 2021a). We used a custom dataset, ANet-HL, derived from temporal localization data. We extract video segments between the start and end times of the target's appearance and use CLIP to calculate the similarity between each frame's scene and the target. This is converted into discrete saliency levels ranging from 1 to 5, at intervals of 0.5. This task effectively enhances the model's temporal perception capabilities for specific events.

Temporal Grounded Caption. This task involves using scene titles as queries, requiring the model to output both the time segments when the scenes appear and the fine-grained subtitles for those segments. We used our custom dataset, CosMo-TGC. This task format, which combines temporal localization and semantic understanding, can effectively prevent large language models from focusing on irrelevant video segments, thereby improving the quality of the model's responses to questions.

We also used normal data comprising four tasks and six different data sources. These general data help prevent the model from overfitting to temporal grounding-related tasks during training, thereby preserving the model's original capabilities.

C Computational Efficiency

By applying Token Shuffle, we further reduced the computational cost of VideoChat-T, giving it a significant computational advantage over high-performance models like LLaVA-OneVision (Li et al., 2024a) and Qwen2-VL (Wang et al., 2024a). Under the same settings, VideoChat-T uses only 3 tokens per frame, with flops consumption at just 5.1% of LLaVA-OneVision. Its inference time on single A100 is only 0.63 seconds, reaching real-time response levels, making it highly suitable for applications requiring rapid response, such as online video understanding.

Method	Token num per frame	flops 128 frames	Inference Time 128f & on single A100 GPU	Charades-STA IOU0.5	QVHighlight mAP	MVBench Avg	Egoschema Full	VideoMME Vision
Qwen2-VL (Wang et al., 2024a)	138	929.8 T	Out Of Memory	15.0	13.0	67.0	66.7	63.3
LLaVA-OneVision (Li et al., 2024a)	196	693.7 T	4.95 s	7.3	14.98	56.7	60.1	58.2
VideoChat-T (Ours)	3	35.5 T	0.63 s	48.7	26.5	59.9	60.0	46.3

Table 7: Comparison of the computational efficiency and performance of VideoChat-T with other methods. Our approach achieves relatively impressive performance with extremely low computational cost.

In terms of performance, VideoChat-T significantly outperforms LLaVA-OneVision in temporal grounding tasks. It has a slight advantage on MVBench; both perform comparably on Egoschema; but VideoChat-T performs worse on VideoMME. Given the substantial savings in computational resources with VideoChat-T, we consider the disadvantages on some datasets to be acceptable.

Moreover, our model's ability to maintain reasonable performance under high compression ratios suggests that the token embedding spaces of contemporary models may be characterized by considerable feature redundancy. This observation presents a promising avenue for future research, as efficient techniques for compressing or discarding redundant features could substantially reduce computational costs without sacrificing model performance, enabling longer context reasoning.

D Details of Hyperparameters

config	epoch1	epoch2&3
input frame	192	128
max text length	1536	1024
freeze TAPE	True	False
learning rate	2e-5	1.5e-5
input resolution	224	
clip frame	8	
merge lenth	4	
QFormer token (per clip)	96	
lora rank	16	
lora alpha	32	
lora dropout	0.1	
batch size (per GPU)	2	
optimizer	AdamW	
optimizer momentum	0.9, 0.999	
weight decay	0.02	
learning rate schedule	cosine decay	

Table 8: Hyper-parameter Settings During the Training Process of VideoChat-T.

Table 8 lists the hyperparameters used during different epochs of the training process. In the first epoch, we used a larger number of input frames and froze the TAPE. At the beginning of the second epoch, we unfroze the TAPE and fixed the model's input frames to 128. Following the settings of VideoChat2, we integrated the lora module into the LLM and applied flash attention to accelerate the training process.

E Full performances

Model	LLM	Avg	AS	AP	AA	FA	UA	OE	OI	OS	MD	AL	ST	AC	MC	MA	SC	FP	CO	EN	ER	CI
VideoChatGPT (Maaz et al., 2023)	7B	**32.7**	23.5	26.0	62.0	22.5	26.5	54.0	28.0	40.0	23.0	20.0	31.0	30.5	25.5	39.5	48.5	29.0	33.0	29.5	26.0	35.5
VideoLLaMA (Zhang et al., 2023)	7B	**34.1**	27.5	25.5	51.0	29.0	39.0	48.0	40.5	38.0	22.5	22.5	43.0	34.0	22.5	32.5	45.5	32.5	40.0	30.0	21.0	37.0
VideoChat (Li et al., 2023b)	7B	**35.5**	33.5	26.5	56.0	33.5	40.5	53.0	40.5	30.0	25.5	27.0	48.5	35.0	20.5	42.5	46.0	26.5	41.0	23.5	23.5	36.0
ST-LLM (Liu et al., 2024b)	7B	**54.9**	66.0	53.5	84.0	44.0	58.5	80.5	73.5	38.5	42.5	31.0	86.5	36.5	56.5	78.5	43.0	44.5	46.5	34.5	41.5	58.5
VideoChat2 (Li et al., 2024b)	7B	**60.4**	75.5	58.0	83.5	50.5	60.5	87.5	74.5	45.0	47.5	44.0	82.5	37.0	64.5	87.5	51.0	66.5	47.0	35.0	37.0	72.5
VideoChat-T	7B	**59.9**	83.5	68.5	80.5	44.0	61.0	71.0	84.0	35.5	48.0	56.5	87.0	46.0	56.5	78.0	49.5	59.0	46.0	37.0	40.0	66.5

Table 9: The full performance of VideoChat-T on MVBench. VideoChat-T still demonstrates strong performance, effectively prevents catastrophic forgetting caused by incremental fine-tuning.